

Integrals

You are expected to have the following facts memorized.

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1)$$

$$2. \int \frac{1}{x} dx = \ln |x| + C$$

$$3. \int e^x dx = e^x + C$$

$$4. \int a^x dx = a^x \frac{1}{\ln a} + C$$

$$5. \int \sin x dx = -\cos x + C,$$

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$$6. \int \tan x dx = \ln |\sec x| + C,$$

$$\int \cot x dx = -\ln |\csc x| + C$$

$$7. \int \sec x dx = \ln |\sec x + \tan x| + C,$$

$$\int \csc x dx = -\ln |\csc x + \cot x| + C$$

$$8. \int \sec^2 x dx = \tan x + C,$$

$$\int \csc^2 x dx = -\cot x + C$$

$$9. \int \sec x \tan x dx = \sec x + C,$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$10. \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$11. \int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$12. \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$13. \int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x + C$$

$$14. \int \sinh x dx = \cosh x + C,$$

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